An Energy-Based JEPA World Model of Drug Perturbations

Predicting single-cell state in latent space on Tahoe-100M

Team Dynamics

HackTheWorld(s) — EB-JEPA

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The premise — JEPA for cell biology

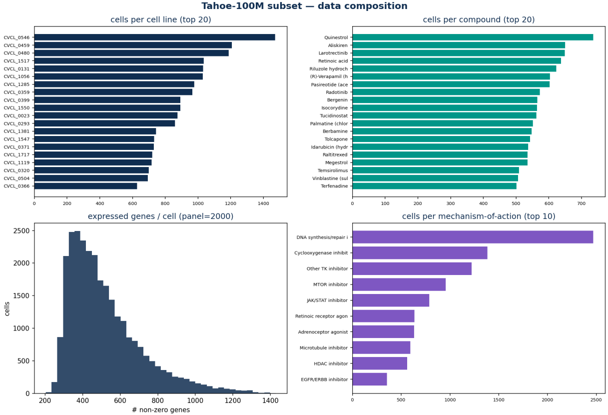
Predict, don't reconstruct

Single-cell transcriptomes are **noisy, sparse, high-dimensional sets** of genes. Reconstructing counts (scGPT) is wasteful; **predicting in a learned latent space** (LeCun's JEPA) discards task-irrelevant noise. *GeneJEPA* (Litman 2025) proved this on Tahoe-100M.

Our goal

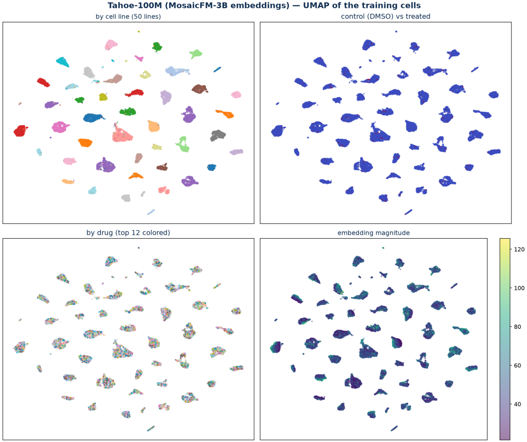
A **true Energy-Based JEPA *world model*** of drug response — **(control cell, drug) $\rightarrow$ perturbed cell** — on a **frozen pretrained encoder**, with **biology-specific losses**, beating the standard baselines.

The data — Tahoe-100M (Vevo/Arc): 100M cells, 1,000 lines, 3,000 drugs



Our subset: a Tahoe-x1 embeddings shard. Each cell = MosaicFM-3B embedding (2560-d) + drug + cell line + MoA. Preprocessing: top-K gene panel, per-feature z-score, co-expression modules.

The data — UMAP of the cell representations



40k cells. Clean structure by **cell line**, a visible **control vs treated** split, and drug-level grouping — a healthy space to build a world model on.

The EB-JEPA recipe in one equation

Energy = prediction error in representation space

$$\mathcal{L} = \underbrace{\left\| g_\phi(f_\theta(x),\, q_\omega(a)) - f_\theta(x') \right\|^2}_{\text{predict target rep. from context + action}} + \lambda\, \mathcal{R}(z)$$

- f_θ **encoder**, g_ϕ **predictor**, q_ω **action encoder**, $\mathcal{R}$ **anti-collapse regularizer**.
- Three settings: image (identity g_ϕ), video (predict next), **action-conditioned (world model)** — ours.

Our model — frozen foundation encoder + learned drug predictor

- f_θ = **frozen** MosaicFM-3B embedding (2560-d) — a pretrained single-cell foundation encoder.
- g_ϕ = GRU predictor, **conditioned on the drug** (action a).
- Energy $= \|g_\phi(z_{\text{ctrl}}, a) - z_{\text{pert}}\|^2$ via `eb_jepa.JEPA.unroll`.
- **Only the predictor is trained.** A drug is a *large*, *controlled* state change $\Rightarrow$ **no collapse** (unlike slow time series).

Pipeline

z_{ctrl} (DMSO cell)
$\downarrow$ $g_\phi(\cdot,$ drug$)$
$\hat{z}_{\text{pert}}$
$\downarrow$ energy vs real z_{pert}

Our contribution — three biology-aware loss terms

Two tracks: **(A) representation** (trainable encoder) = SIGReg + Pathway; **(B) world model** (**frozen** encoder $\Rightarrow$ no collapse, no anti-collapse term) = PerturbationSignature + Pathway.

1. SIGReg (anti-collapse; track A; our choice vs GeneJEPA VICReg)

Forces the latent batch to be **isotropic Gaussian** (Epps–Pulley statistic on random 1-D projections). One hyper-parameter, linear cost, more stable than VICReg's two coefficients.

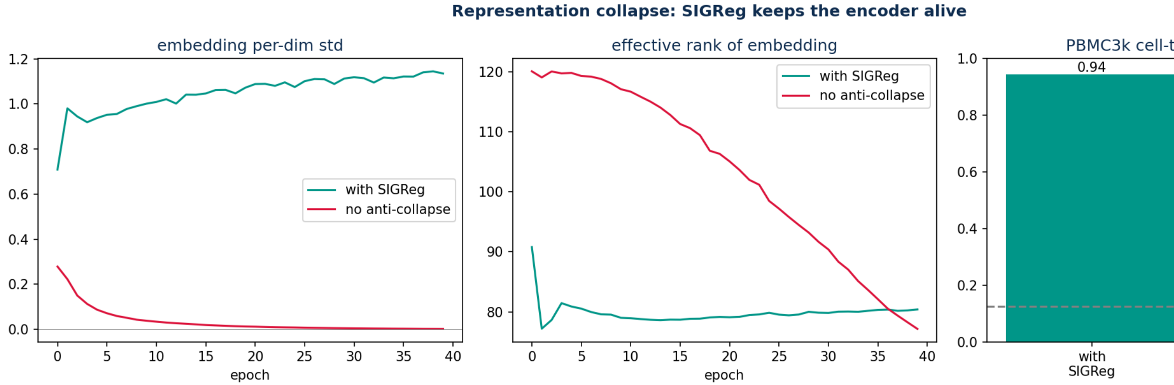
2. PathwayCoherenceLoss (gene-program prior)

Latent pairwise distances must match **co-expression-module activity** distances: $\mathcal{L}_{\text{path}} = \|\widetilde{D}_{\text{latent}} - \widetilde{D}_{\text{module}}\|^2$. Injects biological program structure.

3. PerturbationSignatureLoss (drug consistency)

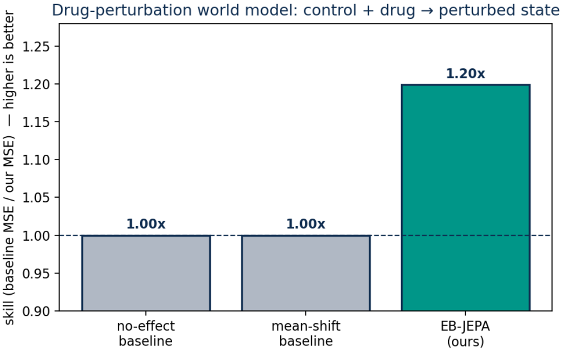
Predicted shift $\hat{z}_{\text{pert}} - z_{\text{ctrl}}$ should point the **same direction for the same drug** (a compound has a characteristic signature) — supervised-contrastive on shift directions.

The collapse we fought — SIGReg keeps the encoder alive



Same cell JEPA on PBMC3k, **with vs without** the anti-collapse term. Without it the embedding **std** $\rightarrow$ 0.002 (collapse), effective rank $\rightarrow 1$, and the cell-type probe drops to **0.43** ($\approx$ chance). With SIGReg: **std** $\approx$ 1.1, **probe** 0.94.

Result — the world model beats both baselines

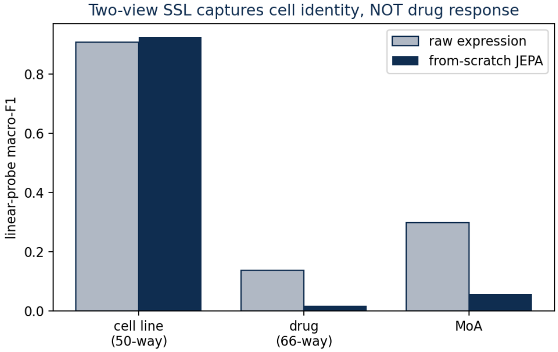


234k train / 59k val cells, real DMSO controls.

predict z_{pert} from...	rel. MSE ↓
no-effect (z_{ctrl})	1.00
mean drug shift	1.00
EB-JEPA (ours)	**0.83**

skill 1.20× vs no-effect, 1.19× vs mean-shift.
Beating *mean-shift* = capturing **context-specific** drug response.

Why a world model? Plain SSL learns identity, not drugs



A from-scratch two-view JEPA on raw genes (700k cells) linear-probes to **F1 0.93 on cell line** but **0.02 on drug**.

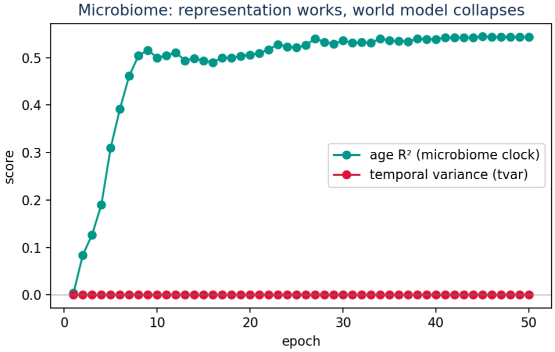
⇒ representation SSL organizes by the **dominant variance (cell identity)**; **drug response needs the action-conditioned predictor**.

Benchmark — PBMC3k cell type (literature dataset)

method	setting	macro-F1
raw expression	in-domain	0.94
PCA-50	in-domain	0.96
our cell JEPA	in-domain	**0.92**
GeneJEPA	frozen transfer	0.69
scGPT	frozen transfer	0.23

Honest read: in-domain, raw expression is near-perfect on PBMC3k (cell types are linearly separable) — this benchmark does *not* discriminate methods in-domain. GeneJEPA's 0.69 is the much harder *frozen-transfer* regime. **Our edge is the perturbation world model, where no trivial baseline exists.**

An honest negative — microbiome dynamics collapse



Same recipe on DIABIMMUNE gut trajectories: representation works (**age** $R^2 \approx 0.53$, microbiome clock) but the world model **collapses temporally** (`tvar`$\to 0$, skill < 1).

Insight: VICReg keeps *batch* variance, not *temporal* variance — slow drift makes "no change" unbeatable. **Controlled perturbations (Tahoe) remove the problem.**

Takeaways

- A **true EB-JEPA world model** of drug perturbation (frozen MosaicFM + learned action predictor) that **beats no-effect & mean-shift baselines ($1.2\times$)** on Tahoe-100M.
- **Three biology-aware design choices**: SIGReg, PathwayCoherenceLoss, PerturbationSignatureLoss — our differentiators from GeneJEPA.
- A **controlled, honest comparison** (+ a clean microbiome collapse finding that motivates the perturbation setting).

Predict cellular response to drugs in latent space — a step toward in-silico screening.